# SAD SOLUTIONS

Systems Analysis & Design — G-QuickLunch Smart School Canteen App


INSTITUTION
**GIMPA Elementary**

AUTHOR
**Sammi Laryea**

STATUS
**Completed**


## Part 1: The Three Feasibility Checks

### Economic Feasibility

**Upfront Cost:** Procurement of hardware equipment, including tablet stations, barcode scanners, and system setup fees.

**Savings/Recovery:** Eliminates continuous recurring costs associated with printing, issuing, and managing physical paper tickets.

### Technical Feasibility

**Required Tool:** Centralized cloud database engine and server architecture connected to school network infrastructure.

**Technical Failure:** Local Wi-Fi connection outage during lunch hours preventing balance sync and transaction authorizations.

### Operational Feasibility

**Staff Acceptance:** Streamlines checkout processes and eliminates manual verification and ticket logging burdens.

**Human Factor Risk:** Students losing or forgetting physical barcode cards, creating manual lookup bottlenecks and queues.

## Part 2: STRIDE Threat Analysis

### Threat 1: Tampering (T)

**Realistic Attack:** A student intercepts network traffic or manipulates tablet client parameters to alter transaction amounts or artificially inflate account balances.

**Concrete Defense:** Enforce HMAC cryptographic message signatures on all API payloads, lock tablets into locked Kiosk Mode, and prevent local settings access.

### Threat 2: Spoofing (S)

**Realistic Attack:** A student borrows or steals another student's plastic barcode card to scan and purchase meals charged to the victim's account.

**Concrete Defense:** Mandate visual facial verification by cafeteria staff using the student photo and name displayed on the tablet before tapping "Approve".

### Threat 3: Elevation of Privilege (E)

**Realistic Attack:** A student bypasses UI restrictions to access kitchen staff controls on the tablet station and self-approves unbilled meal orders.

**Concrete Defense:** Implement Role-Based Access Control (RBAC) requiring staff-authenticated PIN or biometric credentials to enable approval actions.

## Part 3: The Go / No-Go Decision